

Damping of breathing vortices in a dipole pair-plasma condensate

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Two box-scale vortices breathe on the zonal condensate of a strongly driven dipole pair plasma. We show that they are the shear-flow instability of their own jets, saturated by trapping, and that they are damped by a mechanism peculiar to a plasma that is not quasineutral at their scale: Poisson’s equation, with a metric that varies along the magnetic field, forces on every vortex a small part that varies along the field, and the particles streaming along the field Landau damp it. We calculate the damping rate with no adjustable parameter, and it is the measured rate in every simulation, over a fourfold change of the box and of the Debye length. The resonant particles absorb the momentum of the vortices together with their energy, and in the magnetic field the momentum becomes a radial current that drives the jets. The damping is therefore not a friction: it conserves the momentum of the jets and the vortices, and, in the cores of the jets, it passes energy from the jets to the vortices. The jets pay for most of the loss, and the vortices drain only by the small difference between this exchange and their Reynolds stress, far more slowly than they are damped.

1. Introduction

Strongly driven electron–positron plasmas in a dipole collapse, after a single explosive burst of interchange activity, into a box-scale zonal condensate, i.e., a flow that does not vary in the direction of the flow itself and holds nearly all of the energy, on which two coherent vortices ride, one on each jet, carried in opposite directions and exchanging electrostatic energy with the zonal flow at twice their advection frequency — the ‘breathing’ of Kennedy & Plunk (2026*b*). Nothing in the collisionless transit-averaged model damps the flute-like zonal part of this state (flute-like meaning constant along the magnetic field line) (Kennedy & Plunk 2026*a,b*), and the net injection of free energy by the gradients is negligible; the condensate is therefore long-lived, and the transport it carries is close to zero. The vortices, however, are not protected in the same way: they lose energy to the particles at a steady rate, yet they survive for thousands of time units, draining at a small fraction of that rate. Coherent structures riding on sheared flows are common in zonally dominated plasma turbulence (Ivanov *et al.* 2020, 2025), and their lifetime can be as important as their existence. Since the vortices carry the large oscillating fluxes of the condensate, what removes them — and whether it would also remove them in a less idealised plasma — determines the long-time state of the system.

The vortices are the shear-flow instability of their own jets. The zonal flow of the condensate is not a single harmonic: its shear layers are corrugated, on a scale of a few Debye lengths, and such jets are unstable, in the sense of Lord Rayleigh (1880), to

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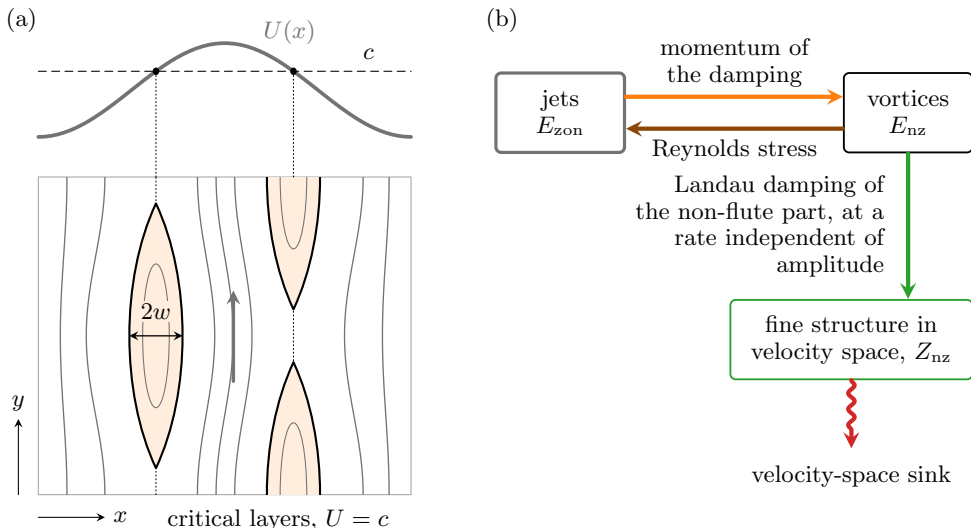


FIGURE 1. A cartoon of how the vortices lose their energy. (a) The geometry: the zonal flow $U(x)$ of one jet (grey, top) and the phase speed c of its vortex (dashed), which cross at two critical layers on the flanks of the jet (dotted); below, the streamlines in the frame of the vortex, with a cat’s eye of trapped fluid, of half-width w , at each critical layer, the two rows of eyes staggered by half a wavelength, and the jet (grey arrow) meandering between them. (b) The flow of energy. The vortices lose electrostatic energy E_{nz} to entropy Z_{nz} , i.e., fine structure in velocity space, by Landau damping of the small part of their potential that varies along the field line (green), at a rate that does not depend on their amplitude, and the velocity-space sink removes it (red). The particles that absorb the energy also absorb the momentum of the vortices and hand it to the jets, which passes energy from the jets to the vortices (orange); the Reynolds stress passes energy back (dark orange). The two exchanges nearly cancel: the jets pay for most of the loss, and the vortices drain by the small difference.

perturbations of the scale of the box. The instability saturates when the fluid at its critical layers, viz., the surfaces on which the flow moves with the perturbation, is trapped in ‘cat’s eyes’ of closed streamlines (O’Neil 1965; Stewartson 1977; Warn & Warn 1978); the saturated state is the pair of breathing vortices, which, in turn, maintain the corrugation of their jets. This state is not lossless. A vortex is flute-like only at scales larger than the Debye length; the small part of its potential that varies along the field line does work on the particles that stream along it, and this Landau damping (Landau 1946) converts the electrostatic energy of the vortex into fine structure in velocity space, at a rate that does not depend on its amplitude. The particles absorb the momentum of the vortex together with its energy, and hand it to the jets. As a result, the jets, not the vortices, pay for most of the loss: the damping and the Reynolds stress move energy between the jets and the vortices in opposite directions, and the vortices drain by the small difference. This picture is sketched in figure 1.

The damping mechanism belongs to a known family. A mode that is two-dimensional to leading order can acquire structure along the magnetic field from the geometry of the confinement, and particles that bounce or transit along the field then damp it at their orbit resonances: the rotational pumping of the diocotron mode of a non-neutral plasma, where the structure comes from the ends of the trap (Crooks & O’Neil 1995), bounce-harmonic Landau damping (Berk & Book 1969; Anderegge *et al.* 2016), and the collisionless damping of the geodesic acoustic mode of a tokamak (Sugama & Watanabe 2006). What is particular here is the origin of the structure: in a plasma that is not

quasineutral at the scale of the vortices, Poisson's equation with a metric that varies along the field line forces a non-flute part of relative size $k^2 \lambda_D^2$ on any vortex, and the vortices on which it acts are self-organised, in jets that Doppler-shift the resonance. That the particles absorb the momentum of the wave together with its energy, and that in a magnetic field the momentum becomes a radial current that drives the mean flow, is likewise known, in the theory of waves and mean flows (Andrews & McIntyre 1976; Bühler 2014), in the momentum theorems for zonal flows (Diamond *et al.* 2008), in wave-driven transport in fusion plasmas (Fisch & Rax 1992) and in the damping of diocotron and Kelvin modes (Crooks & O'Neil 1995; Kabantsev *et al.* 2014); its consequence here, that the damping takes energy from the jets and gives part of it to the vortices, is what keeps them alive.

In this paper, we first summarise the setting in §2. In §3, we derive the theory: the fluid limit and the instability of the jets (§3.1 and §3.2), its saturation by trapping (§3.3), the Landau damping (§3.4), the momentum that it carries (§3.5) and the energy budget that follows (§3.6). In §4, we test each of these against the simulations, and §5 summarises the results.

2. The setting

In this section, we recall the equations of Kennedy & Plunk (2026*b*) and Kennedy & Plunk (2026*a*) to the extent needed here, describe the strongly driven condensate in which the vortices live, and specify the simulations analysed in §4.

2.1. Transit-averaged equations

We consider a symmetric electron–positron plasma, with charges $e_p = -e_e = e$, a common Maxwellian f_0 of density n_0 and temperature T , in the ordering $\rho \ll \lambda_D \ll L$, where ρ is the Larmor radius, $\lambda_D = \sqrt{\epsilon_0 T / 2n_0 e^2}$ the Debye length and L an equilibrium length. We use flux coordinates (ψ, α, l) with $\mathbf{B} = \nabla\psi \times \nabla\alpha$, and write $\delta f_a = h_a - e_a \phi f_0 / T$. Below the transit and bounce frequencies, $h_a(\psi, \alpha, \varepsilon, \lambda, t)$ is independent of l (Helander 2014; Mishchenko *et al.* 2018), where $\varepsilon = mv^2/2$ is the energy and $\lambda = v_\perp^2 / Bv^2$ the pitch-angle variable, and particles respond to the orbit-averaged potential

$$\bar{\phi}(\lambda) = \frac{1}{\hat{\tau}(\lambda)} \int_{l_1}^{l_2} \frac{\phi(l) dl}{\sqrt{1 - \lambda B(l)}}, \quad \hat{\tau}(\lambda) = \int_{l_1}^{l_2} \frac{dl}{\sqrt{1 - \lambda B(l)}}, \quad (2.1)$$

where l_1 and l_2 are the bounce points of a trapped particle or the ends of one field-line period for a passing one. To leading order in $k_\perp \rho$, the kinetic and field equations are (Abel *et al.* 2013; Mishchenko *et al.* 2018; Kennedy & Plunk 2026*a*)

$$\partial_t h_a + \{\bar{\phi}, h_a\} + \bar{v}_{da} \partial_\alpha h_a = \frac{e_a f_0}{T} (\partial_t \bar{\phi} - v_{*a}^T \partial_\alpha \bar{\phi}), \quad (2.2)$$

$$(1 - \lambda_D^2 \nabla_\perp^2) \frac{e\phi}{T} = \frac{1}{2n_0} \int d^3\mathbf{v} (h_p - h_e), \quad (2.3)$$

where $\{X, Y\} \equiv \partial_\psi X \partial_\alpha Y - \partial_\alpha X \partial_\psi Y$ is the $\mathbf{E} \times \mathbf{B}$ advection, $\bar{v}_{da}$ is the orbit-averaged binormal magnetic drift, which is energy dependent and of opposite sign for the two species, and $v_{*a}^T = -(T/e_a) \partial_\psi \ln f_0$ is the gradient drive at fixed energy. The dipole has no radial magnetic drift, so the zonal flow suffers no collisionless damping.

The nonlinearity of (2.2) conserves the free energy W and the electrostatic energy E separately, and their difference $Z = W - E \geq 0$ is the entropy of the orbit-averaged perturbation (Kennedy & Plunk 2026*a, b*); these invariants are the closed-field-line counterparts of those of two-dimensional gyrokinetics (Plunk *et al.* 2010; Plunk &

Tatsuno 2011). Here, we need only two further facts. Firstly, the free energy obeys

$$\frac{dW}{dt} = I - D, \quad (2.4)$$

where I is the injection by the gradients and $D \geq 0$ is the irreversible removal, which, in the simulations, is entirely due to hyperdiffusion (Krommes & Hu 1994; Sugama *et al.* 1996). Secondly, if we split $E = E_{\text{zon}} + E_{\text{nz}}$ into its zonal ($k_y = 0$) and non-zonal parts, the nonlinear transfer between the two satisfies

$$(\partial_t E_{\text{zon}})_{\text{NL}} = -(\partial_t E_{\text{nz}})_{\text{NL}}, \quad (2.5)$$

i.e., the zonal flow and the vortices exchange electrostatic energy without loss; this exchange is the breathing (Kennedy & Plunk 2026b). We shall write Z_{nz} and $W_{\text{nz}} = E_{\text{nz}} + Z_{\text{nz}}$ for the non-zonal parts of Z and W , and $\Gamma_E \equiv -d \ln E_{\text{nz}}/dt$ for the decay rate of the non-zonal electrostatic energy. For a structure of fixed shape, $\Gamma_E = 2\gamma_{\text{amp}}$, where γ_{amp} is the decay rate of its amplitude.

2.2. The strongly driven condensate

At $\eta \equiv L_n/L_T = 1$, the drive is above the fluid-interchange threshold (Mishchenko *et al.* 2018), and the interchange grows at the same rate at every wavelength the box allows. A single $k_y = 1$ streamer then collapses, in one burst, into $k_x = 1$ zonal bands, viz., one jet in each direction across the periodic box, modulated by two box-scale vortices, one on each jet (Kennedy & Plunk 2026b). The two vortices move in opposite directions, related by the symmetry of the pair-plasma equations under exchange of the species together with $\phi \rightarrow -\phi$ and $y \rightarrow -y$, and their relative phase sets the tilt of the flow between them, so the exchange (2.5) runs at $\omega_b = 2|\omega_{\text{adv}}|$, twice the advection frequency of either vortex. Over a breathing cycle, the state changes little; over many cycles, the vortices decay. In what follows, we call $k_y = 1$ the vortex and $k_y = 0$ the jet, and the question of this paper is why the former decays while the latter does not.

2.3. The reduced-box simulation and its restarts

All the data in this paper comes from one nonlinear simulation with the gyrokinetic code GENE (Jenko *et al.* 2000) in the dipole flux tube of Kennedy & Plunk (2026b), at $\eta = 1$, started from noise, in a box a quarter of the production box in each perpendicular direction, together with restarts of it. Its parameters are collected in table 1. There is no collision operator: all irreversible removal is by fourth-order hyperdiffusion, spectral in x and y [rate $\nu_{\perp}(\Delta x k_x/2)^4$, where Δx is the radial grid spacing, and likewise in y], and by finite-difference hyperdiffusion along the field line and in $v_{\parallel}$. One property of the grid will matter in §4.1: the radial grid spacing is comparable with the Debye length, $\lambda_D \approx 1.2 \Delta x$. The run is continued for many thousands of breathing periods after the burst, until long after the vortices have disappeared. Times are in $L_{\text{ref}}/c_{\text{ref}}$, lengths in ρ_{ref} , and energies in the units of the GENE free-energy diagnostics (Bañón Navarro *et al.* 2011, 2016).

Two sets of restarts use the same parameters. The *amplitude tests* restart the original run in the slow phase of the decay with every $k_y \geq 1$ component of the distribution function multiplied by $\epsilon = 0.3$ or 0.03 , the zonal component untouched. The *sink tests* restart it in the accelerating phase with one sink changed: the velocity-space hyperdiffusion divided by four, the perpendicular hyperdiffusion multiplied or divided by four, or the radial resolution doubled. Further runs switch the radial hyperdiffusion off, or change the Debye length.

Quantity	Symbol	Value
Gradient drives	$\omega_n = \omega_T$	2.7816 ($\eta = 1$)
Grid	$n_{x0} \times n_{ky0} \times n_{z0}$	$64 \times 16 \times 32$
Velocity grid	n_v, L_v	84, 4.5
Box	$L_x, k_{y,\min}$	$3689 \rho_{\text{ref}}, 2 \times 10^{-3} \rho_{\text{ref}}^{-1}$
Debye length	λ_D	$70.7 \rho_{\text{ref}} = 1.23 \Delta x$
Perpendicular hyperdiffusion	$\nu_x = \nu_y$	2.4×10^{-3} (spectral, fourth order)
Parallel hyperdiffusion	ν_z	0.05 (fourth order)
Velocity-space hyperdiffusion	ν_v	0.2 (fourth order in $v_{\parallel}$)
Collisions	—	none

TABLE 1. A summary of the parameters of the reduced-box simulation analysed in this paper. The restarts share them, except where stated otherwise.

3. Theory of the vortices

In this section, we derive the three elements of the picture of figure 1, viz., the instability of the jets, its saturation and the kinetic loss, and then put them together.

3.1. The fluid limit

At the scale of the box, the jets and the vortices are flute-like, and the magnetic drift is slow compared with their motion. In this limit, the kinetic problem collapses to a fluid one.

Let us take the charge moment $\sum_a e_a \int d^3 \mathbf{v}$ of (2.2) with $\bar{\phi} = \phi$. The bracket commutes with the velocity integral, the gradient drive cancels between the species because $e_a (e_a f_0 / T) v_{*a}^T = -e_a \partial_\psi f_0$ sums to zero, and the drift term survives only through the even part $h_p + h_e$, since $e_a \bar{v}_{da}$ is the same for both species. Using (2.3) to write the charge moment as $(2n_0 e^2 / T)(1 - \lambda_D^2 \nabla_\perp^2) \phi$, and $\{\phi, \phi\} = 0$, we find[†]

$$\partial_t \nabla_\perp^2 \phi + \{\phi, \nabla_\perp^2 \phi\} = \frac{T}{2n_0 e \lambda_D^2} \partial_\alpha \int d^3 \mathbf{v} \bar{v}_{dp} (h_p + h_e), \quad (3.1)$$

where $\bar{v}_{dp}$ is the positron drift. The right-hand side is the effect of the magnetic drift, which is slow compared with the motion of the vortices relative to the flow of their jets; we neglect it from here on, and confirm in §4.5 that it plays no part in their energy budget. (It does not rely on the transit average either: the charge moment of the kinetic equation with the streaming of particles along the field line retained gives the same equation on averaging along the line, and the streaming adds only the loss of §3.4.) What remains is the two-dimensional Euler equation for the streamfunction ϕ and the vorticity $\nabla_\perp^2 \phi$: the Debye length drops out, the gradients drop out, as they do for a symmetric pair plasma in a straight field (Helander 2014), and the system of jets and vortices is an incompressible inviscid flow. Two properties of this limit matter in what follows. Firstly, multiplying the Euler equation by $-\phi$ and integrating over the box shows that it conserves $\int |\nabla_\perp \phi|^2 / 2$, which is the electrostatic energy E of a flute-like potential in units of $n_0 e^2 \lambda_D^2 / T$: the

[†] Strictly speaking, (3.1) holds as written in a flux tube whose metric is constant along the field line; in the dipole, $\nabla_\perp^2$ depends on l , and a flute-like ϕ satisfies it only on average along the line. This is consistent with the approximate flute-like character of the long-wavelength states (Kennedy & Plunk 2026a). In the flux tube of the simulations, the average is weighted by the Jacobian, and its effect on (3.2) is to replace k^2 in the Laplacian by $k^2 \langle g^{yy} \rangle / \langle g^{xx} \rangle$, where g^{xx} and g^{yy} are the metric coefficients; this correction is of order unity, and the stability of the jets depends on it (§4.3).

vortices and the jets can exchange electrostatic energy, by (2.5), but neither can lose it. Secondly, nothing in it refers to the amplitude of the gradients that drive the turbulence in which the condensate was born: the vortices are a property of the jets.

3.2. The instability of the jets

Let $U \equiv \partial_\psi \Phi$ be the velocity of a zonal flow with potential $\Phi(\psi)$.[†] Linearising the Euler equation about it, with $\phi = \Phi + \varphi(\psi) e^{ik\alpha - i\omega t}$, where real parts are understood, and writing $c \equiv \omega/k$ for the phase speed, gives Rayleigh's equation (Lord Rayleigh 1880; Drazin & Reid 2004),

$$(U - c) (\partial_\psi^2 - k^2) \varphi - U'' \varphi = 0, \quad (3.2)$$

in which U'' is the gradient of the zonal vorticity.[‡] The surfaces on which $U = c$ are the critical layers of the perturbation: there, the perturbation is at rest in the frame of the fluid, (3.2) is singular, and the perturbation exchanges energy with the mean flow. For a perturbation whose growth rate γ_R is small compared with its frequency, the exchange at the critical layers determines the growth rate (Briggs *et al.* 1970),

$$\gamma_R = \pi \frac{\sum_c U_c'' |\varphi_c|^2 / |U_c'|}{\partial_{\omega_r} \text{PV} \int d\psi k U'' |\varphi|^2 / (\omega_r - kU)}, \quad (3.3)$$

where $k > 0$, the sum runs over the critical layers, the subscript c denotes evaluation there, ω_r is the real part of the frequency, and PV denotes the principal value. Whether the perturbation takes energy from the mean flow and grows, or gives energy to it and decays, thus depends on the gradient of the zonal vorticity at its critical layers. This is the content of Rayleigh's theorem: a flow can be unstable only if U'' changes sign. The derivation of (3.2) and (3.3) and the proof of the theorem are standard (Drazin & Reid 2004).

Two cases concern us. A jet that consists of a single harmonic of the box is stable to perturbations whose effective wavenumber along the flow exceeds the wavenumber of the jet, which is the case for all perturbations in the box of the simulations. A jet of the same amplitude whose shear layers are corrugated, so that each is split into strips of vorticity and U'' changes sign close to the critical layers, is unstable. The first is the zonal flow of the condensate as it would be without the vortices; the second is the zonal flow that the simulations have (§4.3).

3.3. Saturation by trapping

A growing vortex changes the trajectories of the fluid at its critical layers, and this is what stops its growth. We use local coordinates (x, y) , radial and binormal, and the $\mathbf{E} \times \mathbf{B}$ streamfunction Ψ .[¶] In the frame $\xi = y - ct$ that moves with the vortex, the flow is steady, and the trajectories are the level lines of the Hamiltonian

$$\dot{x} = -\partial_\xi \mathcal{H}, \quad \dot{\xi} = \partial_x \mathcal{H}, \quad \mathcal{H} = \Psi_{\text{zon}}(x) + \Psi_v(x, \xi) - cx, \quad (3.4)$$

where Ψ_{zon} and Ψ_v are the streamfunctions of the jet and of the vortex. Let x_c be a critical layer, $U(x_c) = c$, with shear $S \equiv U'(x_c)$, and let A be the amplitude of the vortex there.

[†] Not to be confused with the specific flux-tube volume $U = \oint dl/B$ of Kennedy & Plunk (2026b), which does not appear in this paper.

[‡] In the flux tube of the simulations, k^2 in the Laplacian of (3.2) is replaced by $k^2 \langle g^{yy} \rangle / \langle g^{xx} \rangle$, where the angle brackets denote the average along the field line of the metric coefficients (§3.4).

[¶] In GENE units, $\Psi = \phi/C_{xy}$ with C_{xy} the metric factor of the flux tube, x is in ρ_{ref} and t in $L_{\text{ref}}/c_{\text{ref}}$. GENE writes the real field as $\phi = \phi_0 + 2 \text{Re}(\phi_{k_y} e^{ik_y y})$, so the cosine amplitude of a single k_y is $A = 2|\phi_{k_y}|/C_{xy}$.

Expanding (3.4) about x_c gives the Hamiltonian of a pendulum,

$$\mathcal{H} = \text{const} + \frac{S}{2}X^2 - A \cos(k\xi) + \dots, \quad X = x - x_c. \quad (3.5)$$

The fluid within a distance w of the critical layer is trapped: it circulates on closed streamlines, the ‘cat’s eye’ of Kelvin, at the frequency ω_{tr} , where

$$\omega_{\text{tr}} = |k| \sqrt{|S|A}, \quad w = 2 \sqrt{\frac{A}{|S|}}, \quad (3.6)$$

are the frequency of small oscillations about the centre of the eye and the half-width of its separatrix (O’Neil 1965); both scale as $A^{1/2}$. On the two flanks of a jet, the shear has opposite signs, and the eyes are staggered by half a wavelength (see figure 1).

The growth rate (3.3) assumes that the fluid crosses the critical layer on its unperturbed trajectory. Once the fluid there has been trapped and has turned over, i.e., after a time of order $1/\omega_{\text{tr}}$, the vorticity inside the cat’s eye has been rearranged and the exchange with the mean flow ceases (O’Neil 1965; Stewartson 1977; Warn & Warn 1978). An instability that grows at the rate γ_R therefore saturates when the trapping frequency has become a fixed multiple of the growth rate,

$$\omega_{\text{tr}} = C \gamma_R, \quad \text{i.e.,} \quad A = \frac{C^2 \gamma_R^2}{k^2 |S|}, \quad (3.7)$$

where C is a pure number, which this argument does not determine. A vortex that has grown in this way has shaped its jets: at its critical layers, the fluid that it has trapped has been mixed, and the gradient of the zonal vorticity there has been flattened. The vortex thus helps to maintain the corrugation that makes its jets unstable; the two are parts of one state, and (3.7) is the relation between them. If the state changes slowly compared with ω_{tr} and γ_R , the relation is maintained: as the jets lose their instability, the vortices decay, and they disappear when the jets become stable.

3.4. The kinetic loss

A flute-like vortex in the fluid limit conserves its energy. The vortices are damped because they are not quite flute-like, and the calculation of the damping has four steps: the motion of a particle along the field line, its response to a potential that varies along the line, the part of the vortex potential that does so, and the work that this part does.

To lowest order, a particle slides along its field line with ε and λ conserved, at $v_{\parallel} = \pm v \sqrt{1 - \lambda B(l)}$. Passing particles go round the periodic flux tube and trapped ones bounce; either way the motion is periodic, with the orbit frequency $\omega_{\tau} = 2\pi/\tau$, where $\tau = \hat{\tau}/v$ or $2\hat{\tau}/v$ with $\hat{\tau}$ of (2.1). Since $\omega_{\tau} \propto v$, every orbit frequency occurs, and for a thermal particle it is of the order of the transit frequency. With the orbit angle $\theta = \omega_{\tau}t$, the potential that a particle sees is a periodic function of θ , $\varphi(l(\theta)) = \sum_n \varphi_n(\lambda) e^{in\theta}$; the harmonic $n = 0$ is the orbit average $\bar{\phi}$, and for a flute-like potential all the others vanish.

Let the vortex be $\phi = \varphi(x, l) e^{iky - i\omega t}$. The zonal flow carries the particles past it, so at the radius x they see it oscillate at the Doppler-shifted frequency $\tilde{\omega}(x) = \omega - kU(x)$, which vanishes at the critical layers. Following a particle along its orbit, without the slow drift and with the gradient terms, which cancel between the species, the non-adiabatic part of its distribution obeys $(d/dt - i\tilde{\omega})g_a = -i\tilde{\omega}(e_a f_0/T)\varphi(l(t))$, with $d/dt = \omega_{\tau}\partial_{\theta}$. Each harmonic is solved separately,

$$g_a = \frac{e_a f_0}{T} \sum_n \frac{\tilde{\omega}}{\tilde{\omega} - n\omega_{\tau} + i0} \varphi_n e^{in\theta}, \quad (3.8)$$

where causality, i.e., a response built up from the past, gives ω a small positive imaginary part, and $1/(X+i0) = \text{PV}(1/X) - i\pi\delta(X)$. The term $n = 0$ is all that the transit-averaged equations of §2.1 retain. The others are resonant: the particles that complete n orbits in one period of the vortex as they see it, $n\omega_\tau = \tilde{\omega}$, respond a quarter of a period out of phase and absorb energy. This is Landau damping (Landau 1946), with the orbit frequency in place of $k_\parallel v_\parallel$. Integrated over velocities, (3.8) gives the charge $(e/T)\rho[\varphi]$ on the right-hand side of (2.3); for a flute-like potential $\rho[\varphi] = \varphi$, a Boltzmann response that exchanges no energy.

A vortex cannot be exactly flute-like, because the left-hand side of (2.3) is not: in the flux tube, $\nabla_\perp^2 = g^{xx}(l)\partial_x^2 + g^{yy}(l)\partial_y^2$, and the two metric coefficients do not vary in proportion along the line. Write $\varphi = \varphi_0(x) + \varphi_1(x, l)$, with φ_0 the volume-weighted average along the line and φ_1 small. The average of the charge, $-\lambda_D^2(\langle g^{xx} \rangle \partial_x^2 + \langle g^{yy} \rangle \partial_y^2)\phi$, is carried with the flow; this is the Euler equation. Its part that varies along the line is carried with the same flow, and since the average is conserved, $\partial_x^2 \phi$ changes following the flow at $-\langle g^{yy} \rangle / \langle g^{xx} \rangle$ times the rate of $\partial_y^2 \phi$, which for the vortex is $i\tilde{\omega}k^2\varphi_0$. The varying charge therefore changes at the rate $i\tilde{\omega}k^2\lambda_D^2 s(l)\varphi_0$, with

$$s(l) \equiv \langle g^{yy} \rangle \left[\frac{g^{xx}(l)}{\langle g^{xx} \rangle} - \frac{g^{yy}(l)}{\langle g^{yy} \rangle} \right]. \quad (3.9)$$

Such a charge can change only through currents along the line, driven by φ_1 , whose induced charge $\rho[\varphi_1] - \varphi_1$ changes at $-i\tilde{\omega}(\rho[\varphi_1] - \varphi_1)$. Equating the two rates gives $\varphi_1 - \rho[\varphi_1] = k^2\lambda_D^2 s \varphi_0$, i.e.,

$$\varphi_1 = k^2\lambda_D^2 \chi(l; \tilde{\omega}) \varphi_0, \quad \chi - \rho[\chi] = s(l). \quad (3.10)$$

The non-flute part is forced by the flute-like part through the geometry, is of relative size $k^2\lambda_D^2$, and is linear in the amplitude. The equation for χ is linear, along the field line, with the ratio of $\tilde{\omega}$ to the transit frequency as its only parameter; we solve it by expanding χ in Fourier harmonics along the line, with the orbits and their harmonics computed in the magnetic field of the flux tube. Because of the resonant term, χ is complex: φ_1 lags behind its forcing.

Only φ_1 has a parallel electric field, and only the resonant particles absorb its work on average. Multiplying the field equation for φ_1 by φ_1^* , averaging along the line and taking the imaginary part, the power that the two species absorb per unit volume is $P = (n_0 e^2 / T) k^4 \lambda_D^4 |\varphi_0|^2 \mathcal{D}(\tilde{\omega})$, where

$$\mathcal{D}(\tilde{\omega}) \equiv -\tilde{\omega} \langle s(l) \text{Im} \chi(l; \tilde{\omega}) \rangle \geq 0 \quad (3.11)$$

is a function of the Doppler-shifted frequency that the geometry fixes once and for all [see figure 9(a)]. It vanishes as $\tilde{\omega} \rightarrow 0$, where the transit-averaged description applies, peaks where $\tilde{\omega}$ is a few times the transit frequency, and falls at large $\tilde{\omega}$, where few particles are fast enough to resonate. Integrating over the radius and dividing by the electrostatic energy of the vortex gives the loss rate

$$\frac{\mathcal{L}}{E_{\text{nz}}} = 2k^2\lambda_D^2 \frac{k^2 \int dx |\varphi_0|^2 \mathcal{D}(\omega - kU)}{\int dx (\langle g^{xx} \rangle |\partial_x \varphi_0|^2 + \langle g^{yy} \rangle k^2 |\varphi_0|^2)}, \quad (3.12)$$

summed over the two vortices in the numerator and in the denominator. It converts the electrostatic energy of the vortices into entropy, which the sinks then remove, at a rate that does not depend on the sinks.

Equation (3.12) has no adjustable parameter, and it makes three statements that can

be tested. Firstly, the loss rate is independent of the amplitude of the vortices. Secondly, it is of the order of $k^2 \lambda_D^2$ times the transit frequency, so the vortices of a larger box are damped more slowly; at a fixed box it depends on the Debye length both through this factor and through the frequency of the vortices, which enters $\mathcal{D}(\tilde{\omega})$. Thirdly, the lag of the non-flute potential, measured at any radius, on either vortex and at any time, is a single function of the local value of $\tilde{\omega}$.

3.5. The momentum of the damping

The particles that absorb energy from a vortex also absorb its momentum, and they hand it to the jets. In the frame that moves with the vortex at its phase speed ω/k , its potential is steady and the energy of a particle is conserved, so a particle that gains energy $\Delta\epsilon$ and binormal momentum Δp_y in the frame of the box has $\Delta\epsilon = (\omega/k)\Delta p_y$. In the frame of the zonal flow it gains $(\tilde{\omega}/k)\Delta p_y$, which is the absorbed power P ; hence the particles at the radius x take binormal momentum from the vortex at the rate

$$f(x) = \frac{k P(x)}{\tilde{\omega}(x)} \quad (3.13)$$

per unit volume. In a strong magnetic field, the binormal momentum of a guiding centre is tied to its radius, $p_y = q_a Bx$, so the momentum displaces a particle by $\Delta x = \Delta p_y / q_a B$. Electrons and positrons gain momentum in the same direction and are displaced in opposite directions, carrying the radial current $J_x = f/B$, whatever their shares. Its divergence changes the polarisation charge of the zonal flow, i.e., its vorticity, so that

$$\partial_t U = f \quad (3.14)$$

in the fluid units of §3.1. The damping thus conserves the momentum of the jets and the vortices together, and removes their energy at the rate $\int P dx$. In the frame of the box, the vortex loses energy at the rate $(\omega/\tilde{\omega})P$: P to the particles and Uf to the jets. Where $\tilde{\omega}$ and ω have opposite signs, in the cores of the jets that move faster than the vortex, the same absorption gives the vortex energy, $-(\omega/\tilde{\omega})P > 0$, and takes it from the jets: the damping is not a friction.

3.6. The energy budget of the vortices

Since the Euler equation conserves the electrostatic energy, the energy E_{nz} of the vortices changes only by exchange with the jets and by the kinetic loss:

$$\frac{dE_{nz}}{dt} = \mathcal{T} - \mathcal{L}, \quad (3.15)$$

where $\mathcal{T}$ is the transfer of electrostatic energy from the jets to the vortices, averaged over the breathing, and $\mathcal{L}$ is the loss (3.12). The total electrostatic energy of the jets and the vortices falls only through $\mathcal{L}$, so the jets pay for whatever part of the loss the vortices do not drain.

The argument of §3.5 fixes the direction of each part of the transfer. The jets do not depend on y , so, like the equilibrium of the particles, they are steady in the frame that moves with a vortex, and every exchange of energy and binormal momentum with the vortex obeys $\Delta E = c \Delta M$, with $c = \omega/k$:

$$\frac{dE_v}{dt} = c \frac{dM_v}{dt}, \quad (3.16)$$

where E_v and M_v are the energy and the binormal momentum of one vortex. Two processes change M_v . The damping removes momentum at the rate $\int f dx$, and with it

the energy $\int (\omega/\tilde{\omega}) P dx$ in the frame of the box; since $\mathcal{D}$ grows with $|\tilde{\omega}|$, most of the absorption is in the cores of the jets, where $\tilde{\omega}$ and ω have opposite signs, so the damping, in the frame of the box, passes energy from the jets to the vortex. The Reynolds stress moves momentum between the vortex and its jet without loss, and passes energy from the vortex to the jet. Each exchange is of the order of the absorbed power, and the energy of the vortex changes by their difference: the particles hand the momentum that they take from the vortex to the jets, the stress takes it back, and the vortex loses energy only through the mismatch between where the one deposits momentum and where the other removes it.

The theory thus makes the following predictions, which we test in §4: (i) the vortices have the frequency and the radial structure of the unstable mode of Rayleigh’s equation on their jets; (ii) their trapping frequency is a fixed multiple of the growth rate of that mode; (iii) a vortex that is reduced in amplitude on unchanged jets regrows; (iv) the vortices lose electrostatic energy to entropy through the streaming of particles along the field line, at the rate (3.12), which does not depend on their amplitude, on the velocity-space sink or on any adjustable parameter; (v) the jets pay for the part of the loss that the vortices do not drain; and (vi) the vortices drain by the small difference of the two exchanges, far more slowly than they are damped.

4. Comparison with the simulations

We now test the predictions of §3.6 against the simulations (§2.3). All measurements of the potential are made at the outboard mid-plane, and $k_y = 1$ denotes the vortices and $k_y = 0$ the jets. The two counter-propagating vortices are separated by a two-frequency least-squares fit, $a_+ e^{-i\omega_+ t} + a_- e^{-i\omega_- t}$, in windows of about sixteen breathing periods.

4.1. The slow drain and the numerical collapse

In the reference run, the energy E_{nz} of the vortices falls slowly for several thousand time units, then ever faster, until they collapse, while the zonal energy barely changes and the net injection I is negligible [see figure 2]. The slow drain is physical, and the acceleration and the collapse are numerical (see figure 3). Dividing the velocity-space hyperdiffusion by four changes neither the loss rate $\mathcal{L}/E_{nz}$, as befits Landau damping, nor the decay. Multiplying the perpendicular hyperdiffusion by four brings on the collapse at once; dividing it by four, or doubling the radial resolution, removes the collapse, although the vortices go on losing energy to entropy at the same rate. With the radial hyperdiffusion switched off, the vortices drain exactly as in the reference run through the slow phase, never collapse, and drain at a nearly constant rate, a small fraction of their loss rate. The radial hyperdiffusion erodes the corrugation of the jets (§4.3), the cat’s eyes narrow, and the vortices collapse when the half-width w of the eyes reaches one grid cell, which in this simulation is also about one Debye length. The grey band in the figures marks this collapse.

4.2. The vortices and their critical layers

Each vortex propagates in the direction of its own jet, at a fixed fraction of the peak speed of the jet [see figure 4(a)]. It is radially broad — a plateau over its own jet, with maxima on both flanks — and its maxima lie just inside its critical layers, where the shear is near its largest. The streamlines of the moving-frame Hamiltonian (3.4) show the cat’s eyes directly, one on each flank of the jet, staggered by half a wavelength [see figure 4(b)].

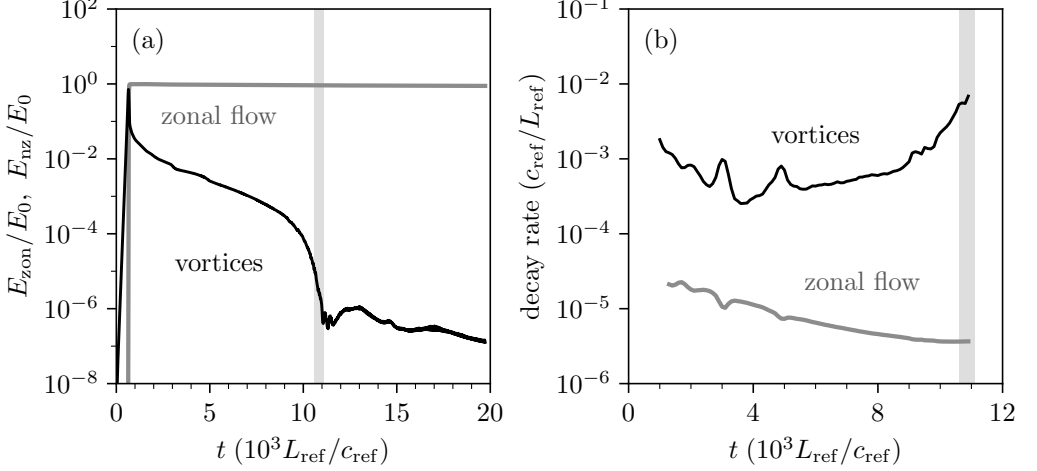


FIGURE 2. Electrostatic energy of the zonal flow and of the vortices in the reduced-box simulation. (a) The zonal energy E_{zon} (grey) and the non-zonal energy E_{nz} (black), as running means over about ten breathing periods, in units of E_0 , the largest value of E_{zon} . (b) The decay rates $\Gamma_E = -d \ln E_{\text{nz}}/dt$ (black) and $-d \ln E_{\text{zon}}/dt$ (grey), from least-squares fits of the logarithm of each energy in sliding windows of about a hundred breathing periods, from the end of the relaxation that follows the burst to the end of the collapse. The grey band marks the collapse in this and all subsequent figures. The vortices lose many decades of energy while the zonal flow barely changes; the late acceleration and the collapse are numerical (figure 3).

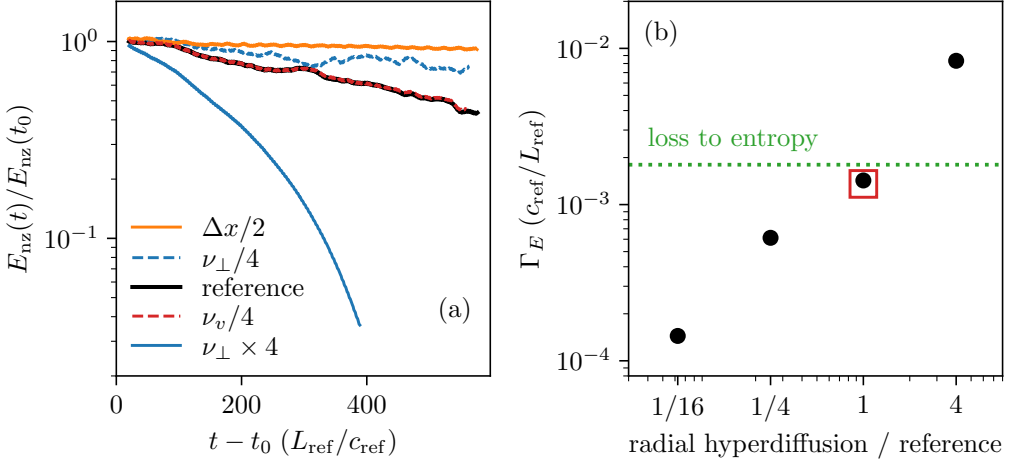


FIGURE 3. The sink tests: restarts of the reference run from one state in the accelerating phase of its decay, at time t_0 , each with one sink changed. (a) The non-zonal electrostatic energy, as running means over about ten breathing periods, for the reference run (black), for the velocity-space hyperdiffusion divided by four (red dashed, on top of the black), for the perpendicular hyperdiffusion multiplied by four (blue solid) and divided by four (blue dashed), and for the radial resolution doubled, which divides the radial hyperdiffusion at a given wavenumber by sixteen (orange). (b) The mean decay rate Γ_E over the interval of (a) against the strength of the radial hyperdiffusion relative to the reference run (black dots; the open red square is the run with the velocity-space hyperdiffusion divided by four), and the loss rate $\mathcal{L}/E_{\text{nz}}$ to entropy, which is the same in all the runs (green dotted). The decay of the vortices follows the perpendicular sink, not the velocity-space one, and where the perpendicular sink is weak the vortices persist although they are Landau damped.

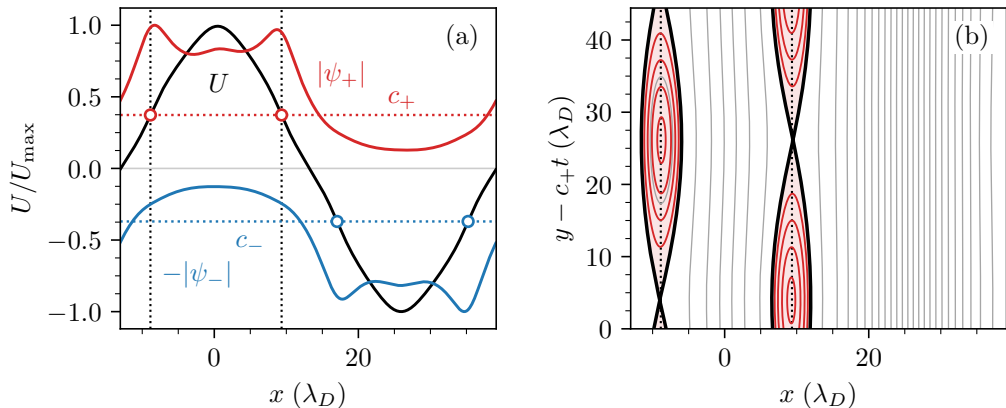


FIGURE 4. Geometry of the vortices on the zonal jets, at one time in the slow phase of the decay. (a) The zonal flow U (black), normalised to its maximum, and the radial profiles of the two vortices, $|\psi_+|$ (red) and $-|\psi_-|$ (blue), each normalised to its own maximum; the horizontal dotted lines are the phase speeds $c_{\pm}$ of the two vortices in the same normalisation, the open circles mark the critical layers $U = c_{\pm}$, and the vertical dotted lines mark those of the $+$ vortex. (b) Streamlines of the moving-frame Hamiltonian (3.4) of the $+$ vortex, with the $-$ vortex left out (grey); the separatrices are drawn in heavy black, and the trapped orbits inside them in red. The radial coordinate is shifted by a quarter of the box so that neither jet is cut by the periodic boundary. Each vortex sits on its own jet, its critical layers lie on the flanks of that jet, not at its maximum, and a cat’s eye several Debye lengths wide forms at each of them.

Two premises of §3 can be checked at once. Firstly, at the wavenumber of the vortices, the drift frequency of a thermal particle[†] is smaller than the frequency of the vortex relative to the flow of its jet by orders of magnitude, and it displaces the resonance of a thermal particle from the critical layer by a small fraction of a grid cell, whereas the cat’s eyes are several grid cells wide. The drift is therefore negligible for the motion of the vortices. Secondly, the breathing modulates the jets by a relative amount of the order of E_{nz}/E_{zon} , which is small, so the jets may be treated as steady in (3.2).

4.3. The vortices are the unstable mode of their jets

Rayleigh’s equation (3.2), with the field-line-averaged metric of the flux tube, has one unstable mode in each direction on the measured zonal flow. Its frequency follows that of the vortices throughout the run, while both change substantially, and its radial structure is that of the vortices [see figure 5(b,c)]. This is prediction (i). The reason for the instability is visible in the zonal vorticity: each shear layer between the jets is split into two strips, and the strips sit on the critical layers of the two vortices, where the gradient of the zonal vorticity is flattened [see figure 5(a)]. This is the mark that the cat’s eyes leave on the jets, and it stays in place throughout the decay. A single harmonic of the same amplitude is stable; it is this corrugation, a few Debye lengths in scale, that makes the jets unstable. The growth rate falls slowly during the decay and vanishes at the collapse [see figure 5(d)].

The corrugation is left by the burst, changes little while the vortices are present, and decays faster once they have gone: the vortices maintain it, and without them the perpendicular sink erodes it. The jets are only just unstable. The growth rate γ_R vanishes at a finite contrast of the strips, not far below that of the slow phase of the decay, so

[†] Here the magnetic drift is the one that the code applies, viz., the curvature and ∇B drift of its geometry, averaged along the field line.

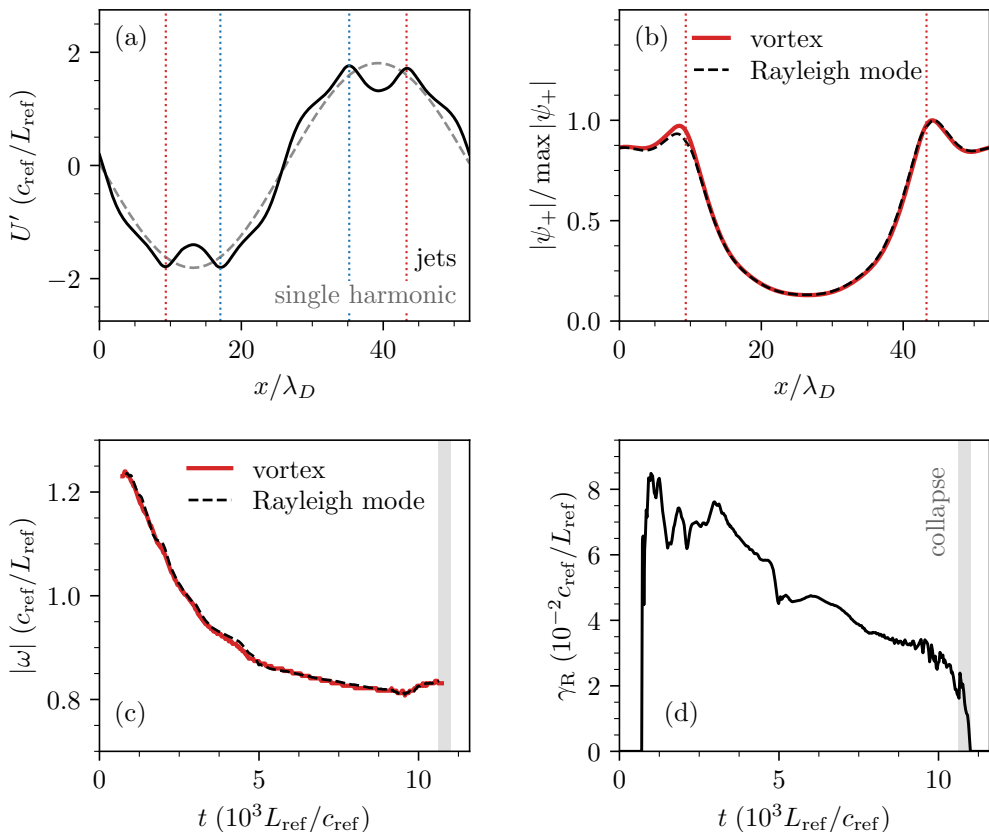


FIGURE 5. The vortices as the shear-flow instability of the jets. (a) The zonal vorticity U' of the measured jets (black) and of a single harmonic of the same amplitude (grey dashed), at one time in the slow phase of the decay; the vertical dotted lines mark the critical layers $U = \pm c$ of the vortex + (red) and - (blue). (b) The radial structure of the vortex + (red) and of the unstable mode of Rayleigh's equation (3.2) on the same jets (black dashed). (c) The frequency of the vortices (red) and of the unstable Rayleigh mode on the measured jets (black dashed) through the run. (d) The growth rate γ_R of that mode; the grey band marks the collapse. The vortices are the unstable shear mode of jets whose shear layers have split into vorticity strips, and they disappear when that instability does.

a small loss of contrast stabilises the jets; this is why the growth rate, and with it the vortices, can decay to nothing while the jets and their strips hardly change.

4.4. The vortices are saturated by trapping

Figure 7 tests prediction (ii), the saturation law (3.7), with ω_{tr} measured from the vortices and γ_R computed from the jets. The law holds through the whole decay of the original run, including the collapse, with a single value of C , while the energy of the vortices falls by orders of magnitude.

The amplitude tests show how the relation is restored when it is broken. Reducing the vortices leaves them below the saturated level of their jets, and they regrow, which is prediction (iii) [see figure 6]. They do not regrow to their former size, however, because, as they regrow, they flatten the corrugation of their jets further and so lower γ_R : the two approach each other. In the $\epsilon = 0.3$ restart, they meet on the curve of the original run.

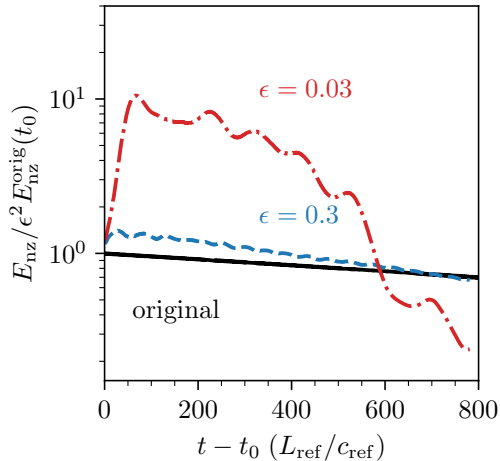


FIGURE 6. The amplitude tests: the non-zonal electrostatic energy of the original run (black solid) and of the restarts with $\epsilon = 0.3$ (blue dashed) and $\epsilon = 0.03$ (red dash-dotted), divided by ϵ^2 and by the energy of the original run at the restart time t_0 , as running means over about ten breathing periods. A vortex that is made smaller on unchanged jets regrows, the more so the smaller it was made.

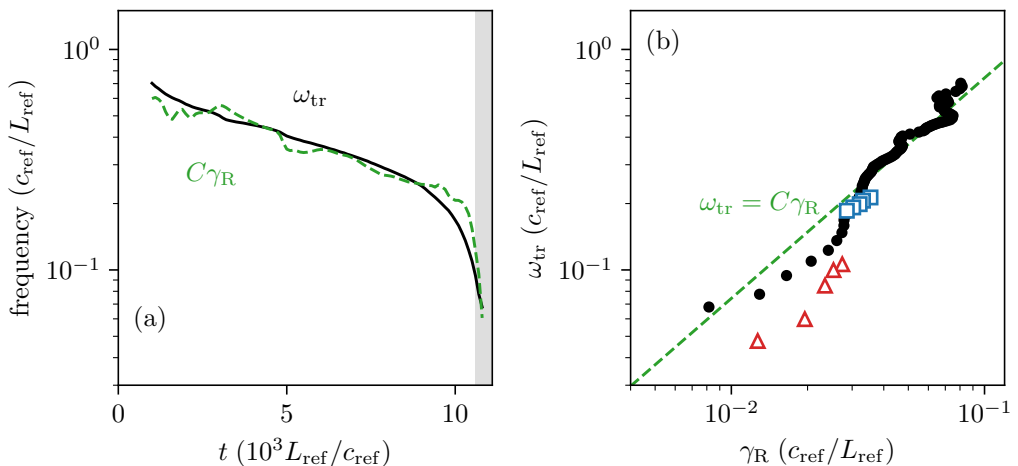


FIGURE 7. The saturation law (3.7). (a) The trapping frequency ω_{tr} of the vortices, from (3.6) at their critical layers (black), and the growth rate γ_R of the unstable Rayleigh mode of the measured jets, multiplied by the constant C (green dashed), both as means over about eighty breathing periods; C is the geometric mean of their ratio, and the grey band marks the collapse. (b) The same data, ω_{tr} against γ_R (black dots), with the line $\omega_{\text{tr}} = C\gamma_R$, and the $\epsilon = 0.3$ (blue squares) and $\epsilon = 0.03$ (red triangles) restarts, with γ_R computed on their own jets. The growth rate is that of the inviscid problem, converged in radial resolution. The trapping frequency of the vortices and the growth rate of the instability of their jets decay together, in a fixed ratio.

In the $\epsilon = 0.03$ restart, whose cat's eyes are no wider than a grid cell, the vortices stay below it [see figure 7(b)].

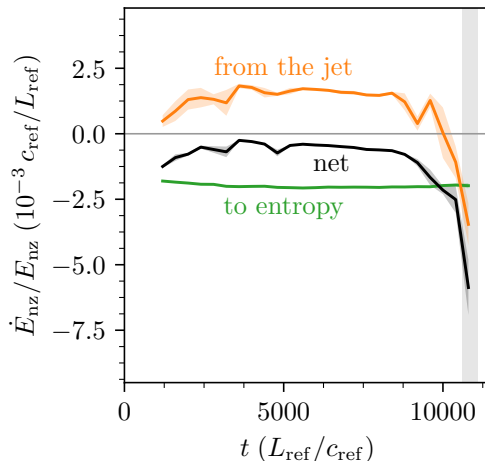


FIGURE 8. The budget (3.15) of the electrostatic energy of the vortices, averaged over the breathing; the rates of change of E_{nz} , per unit E_{nz} , due to the nonlinear exchange with the zonal flow, $\mathcal{T}/E_{nz}$ (orange), due to the linear terms, which convert electrostatic energy into entropy, $-\mathcal{L}/E_{nz}$ (green), and their sum (black); the bands are standard errors over consecutive windows, and the grey band marks the numerical collapse of the reference run. The vortices lose electrostatic energy to entropy at a steady rate, and the jets replace most of it.

4.5. The vortices are Landau damped

The free-energy diagnostics of GENE (Bañón Navarro *et al.* 2011) resolve every term of the energy balance in (k_x, k_y) and separate the electrostatic energy from the entropy, so both terms of the budget (3.15) can be measured (see figure 8).[†] The loss rate $\mathcal{L}/E_{nz}$ is constant while the energy of the vortices falls by orders of magnitude, and it is the same in the amplitude tests as in the original run: the loss is linear in the energy of the vortices. Nearly all of it is carried by the streaming of particles along the field line, and the small remainder by the magnetic drift. The entropy that it produces is removed by the velocity-space hyperdiffusion.

All three statements of §3.4 about the kinetic loss hold (see figure 9). The measured lag follows the predicted $\mathcal{D}(\tilde{\omega})$, with departures close to the critical layers, where $\tilde{\omega}$ is small, and at the largest $\tilde{\omega}$. The loss rate measured by the energy diagnostics of the code, in the channel of parallel streaming, is the predicted one throughout the decay of the reduced-box run, while the energy of the vortices falls by orders of magnitude; it is the same in the restarts with reduced amplitude; and in the production run of Kennedy & Plunk (2026b), whose box is four times larger in each perpendicular direction, it is smaller by the predicted factor. In runs from noise with the Debye length doubled and halved, λ_D^2 changes by a factor of sixteen and the frequency of the vortices by a factor of eight; the measured loss rate changes by only a factor of three, and it is the predicted one in both, with no adjustable parameter. This is the remaining part of prediction (iv). The small non-flute part of the vortices is a single function of the Doppler-shifted frequency for both vortices and at every amplitude, with a peak at the critical layers proportional to their vorticity there and independent of the radial resolution.

[†] All rates are averages over the breathing, which the output resolves with only about three samples per period; the mean is fitted together with two harmonics of ω_b , and the scatter between consecutive windows gives the errors shown in figure 8.

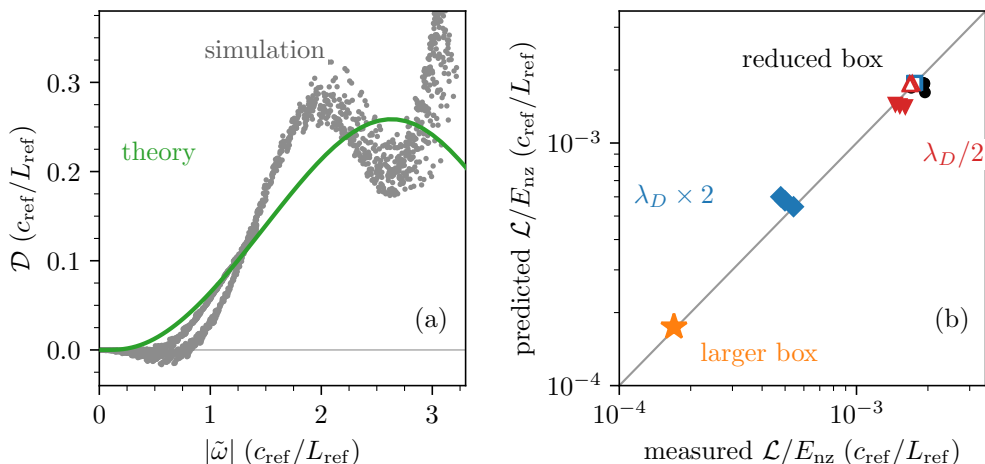


FIGURE 9. Landau damping of the non-flute part of the vortices: theory against simulation. (a) The dissipation function $\mathcal{D}$ of (3.11) against the local Doppler-shifted frequency $|\tilde{\omega}|$: the solution of (3.10) in the magnetic geometry of the flux tube (green line), and its measured counterpart, viz., the part of the non-flute potential that lags behind its forcing, at every radius where either vortex is appreciable and at many times through the decay of the reduced-box run (grey dots). (b) The loss rate $\mathcal{L}/E_{nz}$ predicted by (3.12) from the flute-like part of the vortices and the jets, against the rate measured by the energy diagnostics of the code in the channel of parallel streaming, for the reduced-box run through its decay (black dots), its $\epsilon = 0.3$ (blue squares) and $\epsilon = 0.03$ (red triangles) restarts, and the production run of Kennedy & Plunk (2026b), whose box is four times larger in each perpendicular direction (orange star); the line is equality. The vortices lose their energy by Landau damping of their non-flute part, at the rate that the theory gives, with no adjustable parameter, in every run. Blue diamonds and red downward triangles: runs from noise with the Debye length doubled and halved.

4.6. The jets pay for the loss

The jets replace most of the energy that the vortices lose [see figure 8], and the energy budget of the jets closes: in every run, the rate at which the jets lose energy is the part of the loss of the vortices that the vortices do not drain, plus the hyperdiffusion of the jets, with no adjustable constant [see figure 10]. The Euler Reynolds stress alone does not carry this exchange. Computed from the loss formula, the force (3.13) with which the damping drives the jets has the phase of the part of their nonlinear forcing that the Reynolds stress does not explain, on the dominant harmonic of the jets, and together the two account for that forcing; at the fine harmonics of the jets, the Reynolds stress is the measured forcing.

The two exchanges of §3.6 are each of the order of the absorbed power and of opposite sign for the vortices, and the vortices drain by their small difference, as (3.16) requires, less the energy that they pass to their harmonics [see figure 11]; the harmonics are accounted for below.

The exact budget of the simulation, resolved by binormal wavenumber, shows where the energy goes [see figure 12]. The jets lose energy nonlinearly at about the Landau loss rate of the vortices. Most of it reaches the vortices; the rest reaches their harmonics, the fine binormal structure of the cat's eyes, which are Landau damped as fast as they receive it, at the rate that the loss formula gives at their own wavenumber and Doppler-shifted frequency. The vortices lose a little more through the magnetic drift, and drain by what is left.

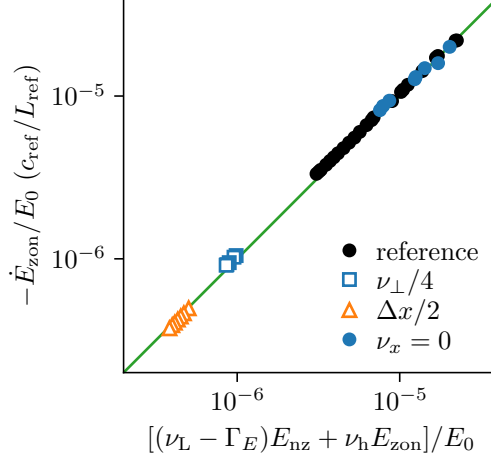


FIGURE 10. How the jets pay. The rate $-\dot{E}_{zon}$ at which the jets lose electrostatic energy, against the part of the kinetic loss of the vortices that they do not drain, $(\nu_L - \Gamma_E)E_{nz}$, plus the radial hyperdiffusion of the jets, $\nu_h E_{zon}$, in windows of about four hundred time units, for the reference run, the restarts with the perpendicular hyperdiffusion divided by four and with doubled radial resolution, and the run without radial hyperdiffusion; the line is equality and there is no adjustable constant. The jets pay for the part of the loss that the vortices do not drain.

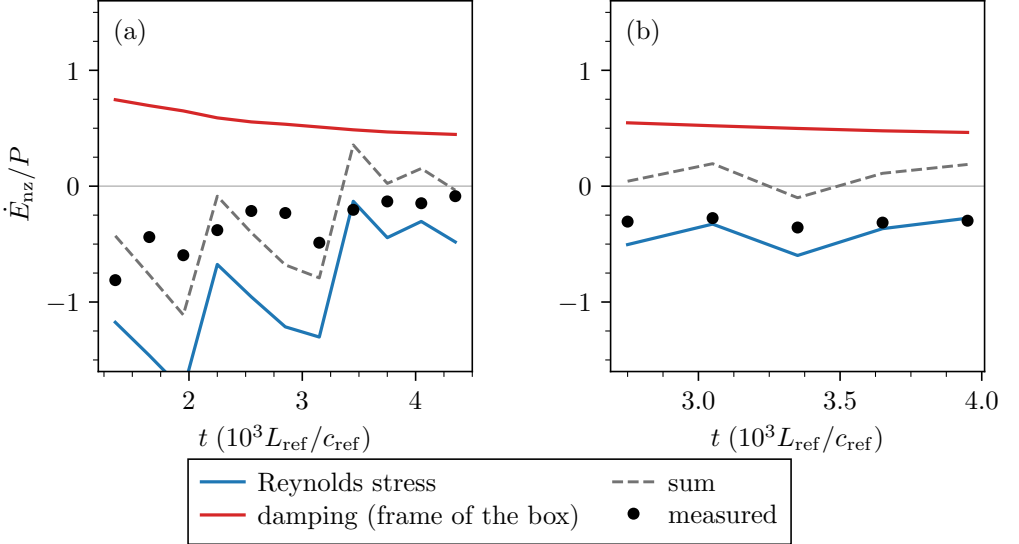


FIGURE 11. The energy of the vortices as the difference of two exchanges with the jets, in the runs without radial hyperdiffusion: (a) at the resolution of the reference run, (b) at doubled radial resolution. The rate at which the vortices gain energy through the Reynolds stress of their field-line-averaged potential (blue) and through their Landau damping in the frame of the box (red), the sum of the two (grey dashed) and the measured rate of change of their energy (dots), all divided by the absorbed power P , in windows of three hundred time units. The two exchanges are each of the order of the absorbed power and of opposite sign; the vortices drain by their small difference, less what they pass to their harmonics (figure 12).

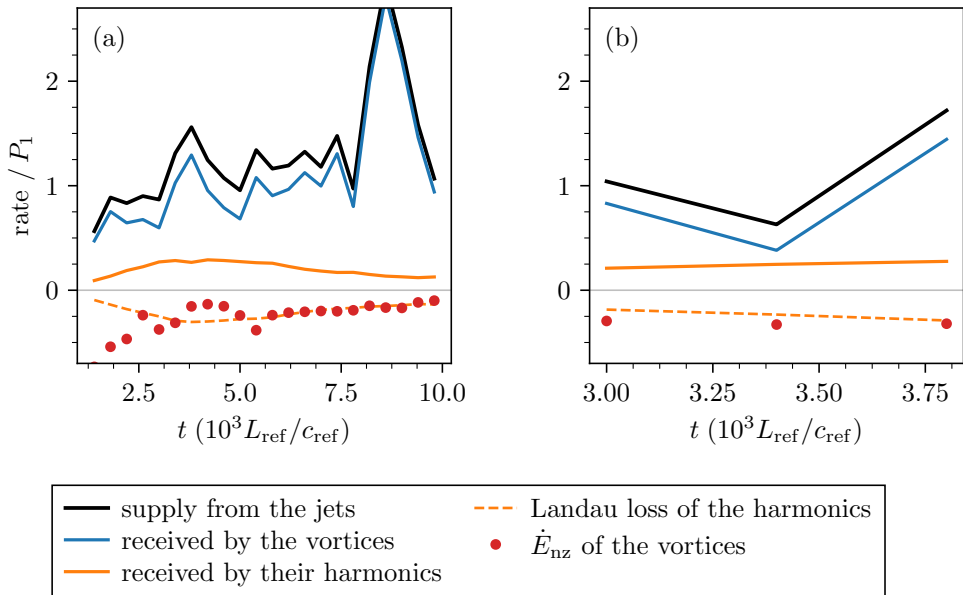


FIGURE 12. Where the energy goes, from the energy budget of the simulation resolved by binormal wavenumber, in the runs without radial hyperdiffusion: (a) at the resolution of the reference run, (b) at doubled radial resolution. All rates are divided by the Landau loss rate P_1 of the vortices: the nonlinear supply from the jets (black), the part of it received by the vortices (blue) and by their harmonics (orange), the Landau loss of the harmonics (orange dashed), and the rate of change of the energy of the vortices (red dots), in windows of eight hundred time units. The jets supply about the Landau loss of the vortices; the vortices pass a small part of what they receive to their harmonics, which lose it at once, and drain by the remainder.

5. Summary and discussion

The breathing vortices of the strongly driven dipole pair-plasma condensate are the shear-flow instability of their own jets. At the scale of the box, the jets and the vortices obey the two-dimensional Euler equation; jets whose shear layers are corrugated on the Debye scale are unstable in the sense of Rayleigh, and the unstable mode on the measured jets has the frequency and the radial structure of the vortices throughout their life. The instability saturates by trapping, and the cat's eyes that it forms maintain the corrugation that drives it.

The vortices are Landau damped by a mechanism that requires a plasma that is not quasineutral at their scale. Poisson's equation, with a metric that varies along the field line, forces on a two-dimensional vortex a part that varies along the line, of relative size $k^2 \lambda_D^2$; the particles that stream along the line at the Doppler-shifted frequency of the vortex absorb it. The loss rate (3.12) has no adjustable parameter, does not depend on the amplitude of the vortices or on the sinks, and is the measured rate in every simulation, over a fourfold change of the box and of the Debye length.

The damping is not a friction. The resonant particles absorb the momentum of the vortices with their energy, and in the magnetic field this momentum becomes a radial current that drives the jets. The damping and the Reynolds stress then move energy between the jets and the vortices in opposite directions, each at about the rate of the loss, and the vortices drain by their small difference. The jets pay for most of the loss, which is why the vortices outlive their damping by far.

The same calculation applies to any two-dimensional structure in a plasma that is not

quasineutral at its scale and whose field lines have a varying metric, such as a non-neutral or pair plasma in a dipole. In such a plasma, without numerical dissipation, the vortices drain slowly; what ends them is whatever erodes the corrugation of their jets, which, in a real plasma, would be collisions.

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Declaration of interests

The authors report no conflict of interest.

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